NETPILOT PROJECT REPORT

Office / Small-Business Network

Design, configuration and testing in Cisco Packet Tracer

Item	Detail
Project owner	**Michel**
Network name	Michel's Office Network
Prepared by	NetPilot
Tool used	Cisco Packet Tracer
Date	October 11, 2026

Figure placeholders. Every dashed box marked **INSERT FIGURE n HERE** shows the exact spot for a Packet Tracer screenshot. Paste the screenshot over the box and keep the caption below it. The List of Figures at the end matches these numbers.

1. Introduction

This report describes the design, configuration and testing of a small-business office network built for Michel. The network connects eight computers in four departments to one router and one switch, and gives them access to the Internet. Each department sits in its own VLAN (virtual LAN) and its own IP subnet, and the router lets the departments exchange messages with each other.

The network uses these technologies:

- **VLANs** to separate departments.
- **802.1Q trunking** to carry all VLANs over one cable between the switch and the router.
- **Inter-VLAN routing** (router-on-a-stick) so departments can communicate.
- **DHCP** so computers can receive addresses automatically.
- **NAT (overload)** so all private addresses share one public address on the Internet.

2. Objectives

1. Build the topology in Packet Tracer with the correct devices and cables.
2. Create one VLAN and subnet per department and explain why each is separate.
3. Give every device an IP address.
4. Make every computer able to send messages to every other computer.
5. Give every computer access to the Internet through NAT.
6. Verify the result with tests and record any problems found and fixed.

3. Network Design

3.1 Why each segment is separate

Segment	Reason for separation
VLAN 10 Executive	Keeps Michel's and management's files and email in a protected group.
VLAN 20 Sales	Handles outside customers and a lot of email, so it is the most exposed to phishing. Separating it limits the damage.
VLAN 30 Accounting	Holds payroll and financial records, so access should be easy to restrict.
VLAN 40 IT	Administrators need tools and access that normal staff should not have.
VLAN 99 Network management	Keeps switch and router admin addresses away from user computers.

Table 1: Network segments and the reason for each.

Separating the network also gives smaller broadcast domains (faster network), easier troubleshooting, and room to grow.

3.2 Topology

The ISP router connects to R1 over the WAN link. R1 connects to the switch SW1 through one trunk link. The eight computers connect to SW1 access ports.

INSERT FIGURE 1 HERE

Screenshot of the complete Packet Tracer workspace showing ISP, R1, SW1 and all eight computers with their names visible.

Figure 1: Network topology of Michel's Office Network in Packet Tracer.

3.3 How a message travels

Inside a VLAN, the switch delivers the message directly from one port to the other using MAC addresses. Between VLANs, the message goes up the trunk to R1, R1 routes it to the destination subnet, and it returns down the same trunk to the correct port.

4. Devices Used

Qty	Device	Packet Tracer model	Name
1	Router (Internet side, for testing)	Cisco 2911	ISP
1	Router (office)	Cisco 2911	R1
1	Switch	Cisco 2960-24TT	SW1
8	Computers	PC-PT (generic PC)	Michel-PC, Exec-Laptop, Sales-PC1, Sales-PC2, Acct-PC1, Acct-PC2, IT-PC1, IT-PC2

Table 2: Devices used in the project.

INSERT FIGURE 2 HERE

Screenshot of the Packet Tracer device palette or the placed devices, showing the router, switch and PC models used.

Figure 2: Devices placed in the workspace.

5. Cabling

From	Port	To	Port	Cable
ISP	Gi0/0	R1	Gi0/1	Copper crossover
R1	Gi0/0	SW1	Gi0/1	Copper straight-through
SW1	Fa0/1	Michel-PC	FastEthernet0	Copper straight-through
SW1	Fa0/2	Exec-Laptop	FastEthernet0	Copper straight-through
SW1	Fa0/5	Sales-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/6	Sales-PC2	FastEthernet0	Copper straight-through
SW1	Fa0/9	Acct-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/10	Acct-PC2	FastEthernet0	Copper straight-through
SW1	Fa0/13	IT-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/14	IT-PC2	FastEthernet0	Copper straight-through

Table 3: Cable and port plan (10 connections: 9 straight-through, 1 crossover).

INSERT FIGURE 3 HERE

Close-up screenshot of the switch SW1 and its cables with all link lights green.

Figure 3: Cabling between SW1, R1 and the computers with all links up.

6. IP Addressing Plan

Each department has one /24 subnet (mask 255.255.255.0). The router is always address .1. Addresses .1 to .50 are reserved for infrastructure and fixed computers, and DHCP hands out .51 to .254.

VLAN	Name	Network	Gateway (R1)
10	Executive	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1
30	Accounting	192.168.30.0/24	192.168.30.1
40	IT	192.168.40.0/24	192.168.40.1
99	Network management	192.168.99.0/24	192.168.99.1

Table 4: VLANs and subnets.

Device	Interface	IP address	Mask	Gateway
ISP	Gi0/0	203.0.113.1	255.255.255.252	-
ISP	Loopback0 (test)	8.8.8.8	255.255.255.255	-
R1	Gi0/1 (WAN)	203.0.113.2	255.255.255.252	203.0.113.1 (default route)
R1	Gi0/0.10 / .20 / .30 / .40 / .99	192.168.10.1 / 20.1 / 30.1 / 40.1 / 99.1	255.255.255.0	-
SW1	Vlan99	192.168.99.2	255.255.255.0	192.168.99.1
Michel-PC	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.1
Exec-Laptop	FastEthernet0	192.168.10.12	255.255.255.0	192.168.10.1
Sales-PC1	FastEthernet0	192.168.20.11	255.255.255.0	192.168.20.1
Sales-PC2	FastEthernet0	192.168.20.12	255.255.255.0	192.168.20.1
Acct-PC1	FastEthernet0	192.168.30.11	255.255.255.0	192.168.30.1
Acct-PC2	FastEthernet0	192.168.30.12	255.255.255.0	192.168.30.1
IT-PC1	FastEthernet0	192.168.40.11	255.255.255.0	192.168.40.1
IT-PC2	FastEthernet0	192.168.40.12	255.255.255.0	192.168.40.1

Table 5: IP addresses of every device. All computers use DNS 8.8.8.8.

7. Configuration

7.1 ISP router

The ISP router was given the WAN address 203.0.113.1/30 on Gi0/0 and a loopback interface with address 8.8.8.8 so the office has something to reach on the "Internet".

INSERT FIGURE 4 HERE

Screenshot of the ISP CLI showing the output of show ip interface brief (Gi0/0 = 203.0.113.1, Loopback0 = 8.8.8.8).

Figure 4: ISP interface configuration.

7.2 Office router R1

R1 was configured with:

- One sub-interface per VLAN on Gi0/0 (Gi0/0.10, .20, .30, .40, .99), each using encapsulation dot1Q and the .1 gateway address.
- The WAN interface Gi0/1 with 203.0.113.2/30 and a default route to 203.0.113.1.
- NAT overload using access list 1 for all 192.168.0.0/16 addresses.
- Four DHCP pools (Executive, Sales, Accounting, IT) with addresses .1 to .50 excluded.

INSERT FIGURE 5 HERE

Screenshot of R1 show ip interface brief showing Gi0/1 = 203.0.113.2 and all five sub-interfaces up/up.

Figure 5: R1 interface status.

INSERT FIGURE 6 HERE

Screenshot of R1 show ip route (connected networks and the default route) or of the DHCP and NAT configuration.

Figure 6: R1 routing table and NAT / DHCP configuration.

7.3 Switch SW1

SW1 was configured with five VLANs (10, 20, 30, 40, 99), access ports for each department, and a trunk on Gi0/1 allowing VLANs 10, 20, 30, 40 and 99. The management address 192.168.99.2 was set on interface Vlan99 with default gateway 192.168.99.1.

VLAN	Name	Ports
10	Executive	Fa0/1 - Fa0/4
20	Sales	Fa0/5 - Fa0/8
30	Accounting	Fa0/9 - Fa0/12
40	IT	Fa0/13 - Fa0/16
99	NetMgmt	none (management interface only)

Table 6: VLAN to switch port assignment.

INSERT FIGURE 7 HERE

Screenshot of SW1 show vlan brief.

Figure 7: VLANs and their ports on SW1.

INSERT FIGURE 8 HERE

Screenshot of SW1 show interfaces trunk showing Gi0/1 trunking with VLANs 10, 20, 30, 40, 99.

Figure 8: Trunk link between SW1 and R1.

7.4 Computers

Each computer was given a static IP address, mask, gateway and DNS server from Table 5 using Desktop > IP Configuration.

INSERT FIGURE 9 HERE

Screenshot of the IP Configuration window of Michel-PC (192.168.10.11, 255.255.255.0, 192.168.10.1, 8.8.8.8).

Figure 9: IP configuration of Michel-PC (the owner's computer).

8. Testing and Results

8.1 Results recorded so far

Test	From	To	Result
ISP and R1 communicate	R1	ISP 203.0.113.1	Success
Own gateway	Michel-PC	192.168.10.1	4 of 4 replies
Internet through NAT	Michel-PC	8.8.8.8	3 of 3 replies
Switch to router	SW1	192.168.99.1	4 of 5 replies (first packet lost while learning the address, which is normal)
Switch to Sales gateway	SW1	192.168.20.1	5 of 5 replies

Table 7: Test results confirmed in Packet Tracer.

INSERT FIGURE 10 HERE

Screenshot of the Michel-PC Command Prompt showing the successful pings to 192.168.10.1 and 8.8.8.8.

Figure 10: Michel-PC reaching its gateway and the Internet.

8.2 Communication between all computers

To confirm that every computer can send messages to every other computer, run the ping tests below and record the result in the last column.

From	To	Path	Result
Michel-PC	Exec-Laptop (192.168.10.12)	Same VLAN, switch only	
Michel-PC	Sales-PC1 (192.168.20.11)	Via R1	
Michel-PC	Acct-PC1 (192.168.30.11)	Via R1	
Michel-PC	IT-PC1 (192.168.40.11)	Via R1	
Sales-PC1	Acct-PC2 (192.168.30.12)	Via R1	
IT-PC1	Every other computer	Via R1	
Any computer	SW1 (192.168.99.2)	Via R1 to VLAN 99	
Any computer	8.8.8.8	R1 and NAT to ISP	

Table 8: Inter-department communication test plan and results.

INSERT FIGURE 11 HERE

Screenshot of a Command Prompt showing successful pings from one department to another (for example Sales-PC1 to Acct-PC1).

Figure 11: Successful communication between two different VLANs through R1.

INSERT FIGURE 12 HERE

Screenshot of Packet Tracer Simulation mode showing a Simple PDU (envelope) message sent from Michel-PC to another computer with the status Successful.

Figure 12: A message sent directly between two computers in Packet Tracer.

9. Problem Found and Solution

Problem. Sales-PC1 (192.168.20.11) could not reach its gateway 192.168.20.1, and Michel-PC could not reach Sales-PC1, even though Michel-PC could reach the Internet. R1 and SW1 were able to reach the 192.168.20.1 gateway, so routing and the trunk were working.

Cause. The show interfaces status output on SW1 showed eight computers connected to Fa0/1 to Fa0/8 and nothing connected to Fa0/9 to Fa0/16. Some computers were cabled to ports belonging to a different VLAN than the one their IP address expects, so their traffic never reached the correct gateway.

Solution. Move each computer to its planned port (Table 3), or change the switch port VLAN assignments to match the cables. Then check again with show interfaces status and repeat the ping tests.

INSERT FIGURE 13 HERE

Screenshot of SW1 show interfaces status showing the problem (devices on the wrong ports, Fa0/9 to Fa0/16 not connected).

Figure 13: SW1 port status before the fix.

INSERT FIGURE 14 HERE

Screenshot of SW1 show interfaces status after the fix, with each computer on its planned port and VLAN.

Figure 14: SW1 port status after the fix.

10. Conclusion

The project built a segmented office network for Michel with separate VLANs for Executive, Sales, Accounting and IT, a management VLAN for the network devices, inter-VLAN routing on R1, DHCP, and NAT for Internet access. The ISP router, R1 and SW1 communicate with each other, and Michel-PC reaches both its gateway and the Internet. The cabling problem found during testing showed the importance of checking switch ports against the design. Once every computer is on its planned port, all departments can exchange messages directly with each other.

11. Recommendations

- Set an enable secret password on R1, SW1 and the ISP router, and use SSH instead of console-only access.
- Add access control lists on R1 to restrict sensitive departments, for example blocking Sales from Accounting and the management VLAN.
- Shut down unused switch ports and move them to an unused VLAN.
- Add a guest Wi-Fi VLAN for visitors.
- Save the configurations (copy running-config startup-config) on every device after each change.

List of Figures

Figure	Description	Section
1	Network topology	3.2
2	Devices placed in the workspace	4
3	Cabling with link lights green	5
4	ISP interface configuration	7.1
5	R1 interface status	7.2
6	R1 routing table and NAT / DHCP	7.2
7	SW1 VLANs and ports	7.3
8	SW1 trunk to R1	7.3
9	Michel-PC IP configuration	7.4
10	Michel-PC ping to gateway and Internet	8.1
11	Ping between two VLANs	8.2
12	Simple PDU message in Simulation mode	8.2
13	SW1 port status before the fix	9
14	SW1 port status after the fix	9

Sign-off

Project owner	Michel	Signature	
Prepared by	NetPilot	Date	October 11, 2026